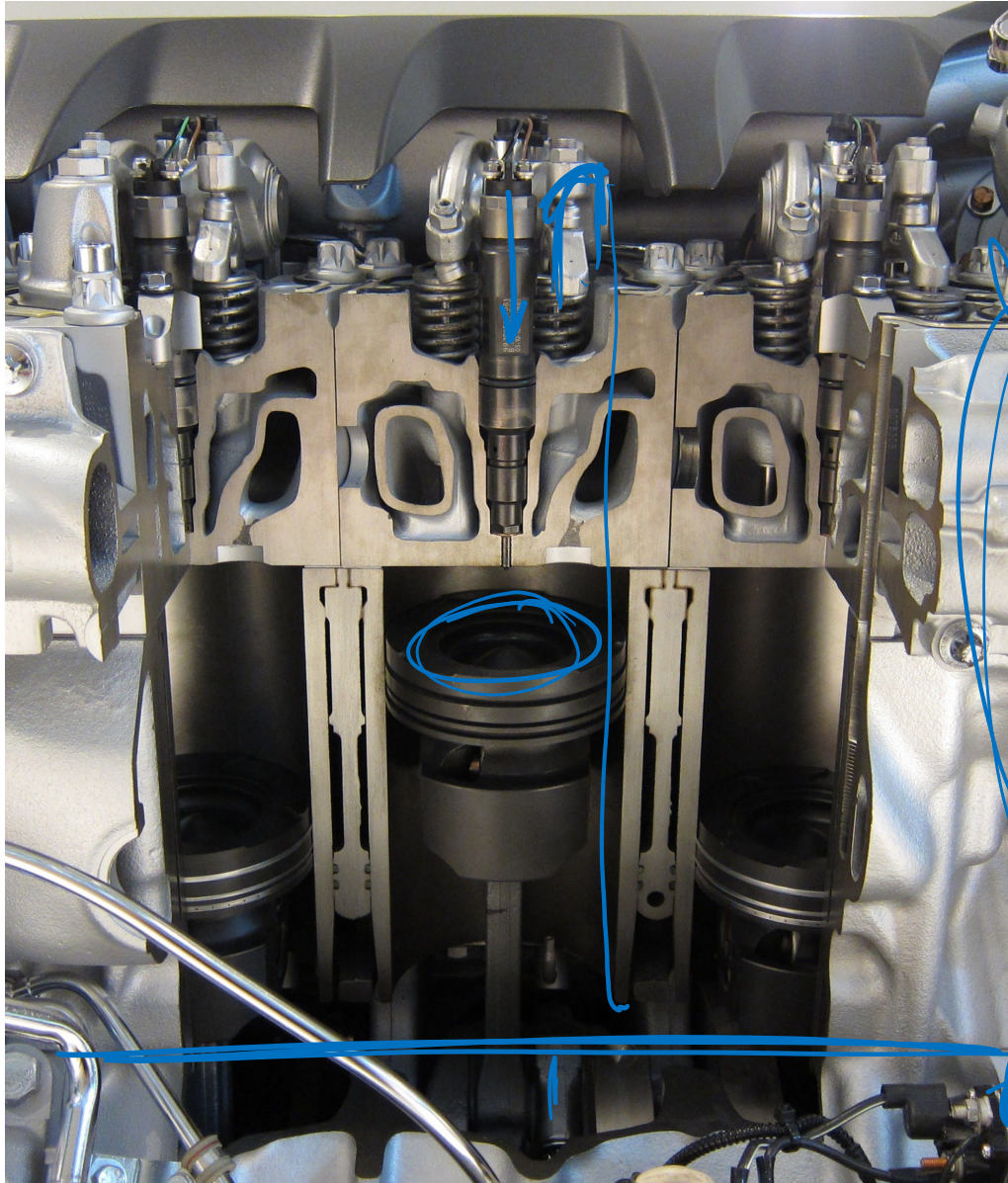




ME 300: Engineering Thermodynamics

Diesel Cycle

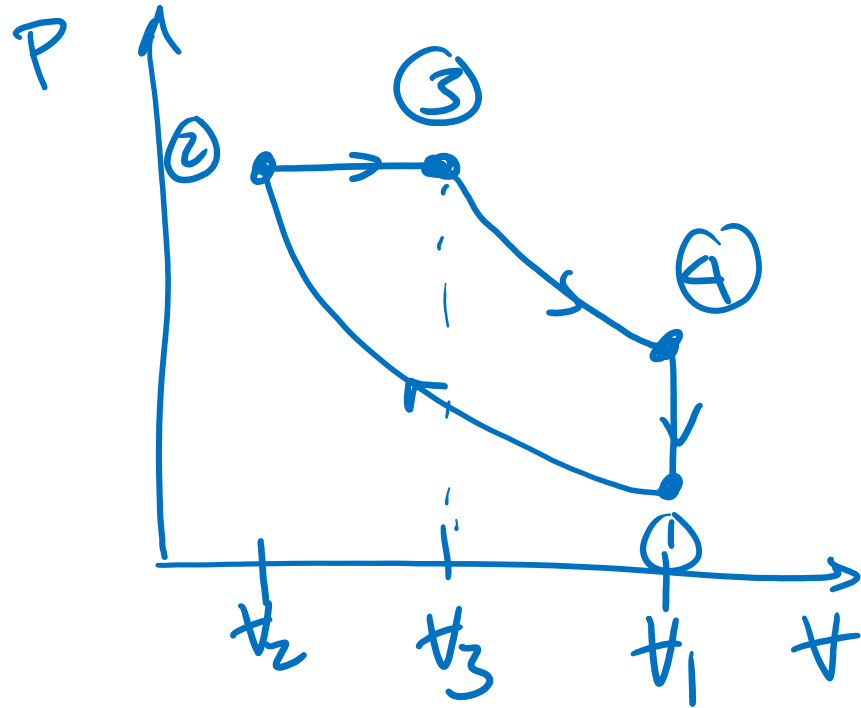
Diesel cycle



Ideal

- 1-2: isentropic compression
- 2-3: isobaric heat addition $\Delta p = 0$
- 3-4: isentropic expansion
- 4-1: isochoric heat rejection $\Delta V = 0$

Diesel cycle



$$r = \frac{v_1}{v_2} \quad d = v_1 - v_2$$

$$\text{Cutoff ratio: } \alpha = \frac{v_3}{v_2}$$

$$\eta_{th} = \frac{\bar{W}_{net}}{Q_{in}} = \frac{{}_1\bar{W}_2 + {}_2\bar{W}_3 + {}_3\bar{W}_4}{{}_2Q_3}$$

$$\eta_{th} = 1 - \frac{1}{r^{\gamma-1}} \left(\frac{\alpha^{\gamma} - 1}{\gamma(\alpha - 1)} \right)$$



For a given compression ratio, which cycle has a higher efficiency?

A. Otto

B. Diesel

